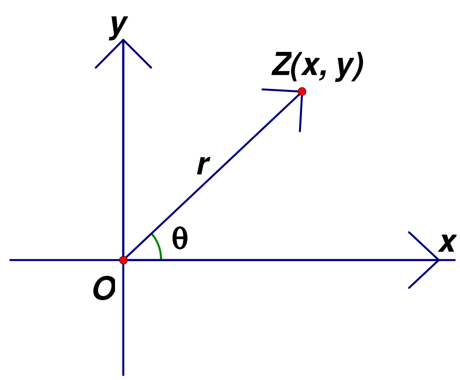
Complex Numbers

Methods of Representations

A *complex* number is a number of the form $z = x + iy$, (*Cartesian form*) with $x, y \in \mathbb{R}$ and $i^2 = -1$, x is the *real part* of z, denoted as $x = \mathrm{Re}(z)$, y is the *imaginary part* of z, denoted as $y = \mathrm{Im}(z)$. Using (x, y) in the xy-plane to represent $z = x + iy$, we observe that $z = r(\cos\theta + i\sin\theta)$, with $r \geq 0$ and $\theta \in \mathbb{R}$, r is the *modulus* of z, denoted as $r = |z|$, θ the *argument* of z, denoted as $\theta = \arg z$. This is the *polar form* of z. A complex number z can also be represented in *exponential form* $z = re^{i\theta}$, (Euler Formula). The three forms are inter-changeable, with $r = \sqrt{x^2 + y^2}$, $x = r\cos\theta$ and $y = r\sin\theta$.



Geometrically a complex number may be regarded as the vector $\overrightarrow{OZ}$.
Additions and subtractions of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are defined component-wise:
$z_1 \pm z_2 = (x_1 \pm x_2) + i(y_1 \pm y_2)$. Geometrically the sum and difference of the two complex numbers correspond to the two diagonals of the parallelogram formed by Z_1 and Z_2.
Multiplications are defined using the distributive property:
$z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = x_1x_2 + ix_1y_2 + ix_2y_1 + i^2y_1y_2 = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$.
In polar forms, using angle formulae, it can be shown that:
$z_1 z_2 = r_1(\cos\theta_1 + i\sin\theta_1)r_2(\cos\theta_2 + i\sin\theta_2) = r_1r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$,
i.e., scaling of magnitude and rotation.
Likewise divisions:

$$\frac{z_1}{z_2} = \frac{r_1(\cos\theta_1 + i\sin\theta_1)}{r_2(\cos\theta_2 + i\sin\theta_2)} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)).$$

Or in Cartesian form $\frac{1}{z}=\frac{1}{x+iy}=\frac{x-iy}{x^2+y^2}$, where $\bar{z}=x-iy$ is the *conjugate* of *z*, so divisions by nonzero complex numbers $(\neq 0=0+i0)$ are possible.

<u>Properties of Complex Conjugates and Modulus</u>

1. $\overline{z_1 \pm z_2}=\overline{z_1}\pm\overline{z_2}$,

2. $\overline{z_1 z_2}=\overline{z_1}\cdot\overline{z_2}$,

3. $\overline{\left(\frac{z_1}{z_2}\right)}=\frac{\overline{z_1}}{\overline{z_2}}, z_2\neq 0$,

4. $z=\bar{z}\Leftrightarrow z\in\mathbb{R}$,

5. $\operatorname{Re} z=\frac{z+\bar{z}}{2}, \operatorname{Im} z=\frac{z-\bar{z}}{2i}$,

6. $z\cdot\bar{z}=|z|^2=|\bar{z}|^2$,

7. $|z_1 z_2|=|z_1|\cdot|z_2|$,

8. $\left|\frac{z_1}{z_2}\right|=\frac{|z_1|}{|z_2|}, z_2\neq 0$,

9. $||z_1|-|z_2||\leq|z_1\pm z_2|\leq|z_1|+|z_2|$,

10. $|z|\geq\max\{\operatorname{Re}(z),\operatorname{Im}(z)\}$.

<u>Roots of Unity and the De-Moivre's Theorem</u>

For the equation $x^n-1=0$ (n a natural number, $n\geq 2$), all its roots are $\varepsilon_k=\cos\frac{2k\pi}{n}+i\sin\frac{2k\pi}{n}, k=0,1,2,...,n-1$, and these roots are called *roots of unity*.
Using complex multiplications, the roots can also be represented as:
$1,\varepsilon_1,\varepsilon_1^2,...,\varepsilon_1^{n-1}$. (This is called the De-Moivre's theorem.)
Basic properties are:

1. $1+\varepsilon_1+\varepsilon_1^2+...+\varepsilon_1^{n-1}=0, n\geq 0$.

2. For any two roots of unity $\varepsilon_i,\varepsilon_j$, there product is still a root of unity, and $\varepsilon_i\varepsilon_j=\varepsilon_{i+j}$, (when $i+j\geq n$, $\varepsilon_i\varepsilon_j=\varepsilon_k$, where k is the remainder when $i+j$ is divided by n.

3. Let m be an integer, $n\neq 1$, then $1+\varepsilon_1^m+\varepsilon_2^m+...+\varepsilon_{n-1}^m=\begin{cases} n, \ n|m \\ 0, \ n \text{ not divisible by } m \end{cases}$.

4. Because $x^n-1=(x-1)(x-\varepsilon_1)(x-\varepsilon_2)...(x-\varepsilon_{n-1})$, get

$x^{n-1}+x^{n-2}+...+1=(x-\varepsilon_1)(x-\varepsilon_2)...(x-\varepsilon_{n-1})$.

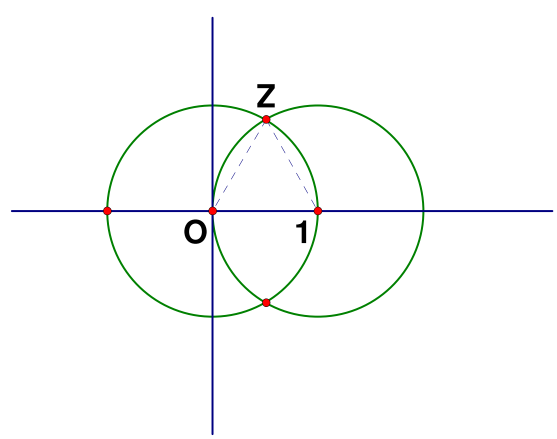
Applications to Algebra

Example: Find all fifth roots of unity.

Use the De-Moivre's theorem all the roots are $\cos\frac{2k\pi}{5}+i\sin\frac{2k\pi}{5}, k=0,1,2,3,4$. Alternately $0=z^5-1=(z-1)(z^4+z^3+z^2+z+1)$. Now $z^4+z^3+z^2+z+1=(z^2+\frac{1}{2}z+1)^2-\frac{5}{4}z^2$. Hence besides the root $z=1$, other roots satisfy $(z^2+\frac{1+\sqrt{5}}{2}z+1)(z^2+\frac{1-\sqrt{5}}{2}z+1)=0$. Therefore we get all roots $1, \frac{-(1+\sqrt{5})\pm i\sqrt{10-2\sqrt{5}}}{4}, \frac{-(1-\sqrt{5})\pm i\sqrt{10+2\sqrt{5}}}{4}$. In particular we have $\cos\frac{2\pi}{5}=\frac{\sqrt{5}-1}{4}$.

Example (CMO 1987, No. 1): Let n be a natural number, show that the equation $z^{n+1}-z^n-1=0$ has a complex root of modulus 1 if and only if $6|n+2$.

If $6|n+2$, let $z=e^{\pm i\frac{\pi}{3}}$, then $z^2=z-1$ and $z^{n+2}=1$. Hence $z=e^{\pm i\frac{\pi}{3}}$ are roots of the equation and they are of modulus 1.



On the other hand, if z is of modulus 1 and is a root of the equation, then $|z|=1$ and $|z^n||z-1|=1$, get $|z-1|=1$. Thus the complex number corresponds to an intersection of the circles $|z|=1$ and $|z-1|=1$. This means $z=e^{\pm i\frac{\pi}{3}}$, hence $z^3=-1, z^6=1$, $z^k=1$ if and only if k is divisible by 6.. Also $z^3+1=0$ and $z\neq -1$, get $z^2-z+1=0$ or $z-1=z^2$, put into the equation, get

$z^{n+2}=1$, thus $6|n+2$.

Example: Show that $\sum_{k\equiv 0(\bmod 3)}\binom{n}{k}=\frac{1}{3}(2^n+2\cos\left(\frac{n\pi}{3}\right))$.

(By the binomial theorem, $(1+x)^n=\sum_{k=0}^{n}\binom{n}{k}x^k$. Put $x=1$, get $2^n=\sum_{k=0}^{n}\binom{n}{k}$. Put $x=-1$, get $0=\sum_{k=0}^{n}\binom{n}{k}(-1)^k$. Hence $\sum_{k\equiv 0(\bmod 2)}\binom{n}{k}=\sum_{k\equiv 1(\bmod 2)}\binom{n}{k}=2^{n-1}$.)

Now let $\omega=\frac{-1+\sqrt{3}}{2}$, then $0=\omega^{3k}-1=(\omega^k-1)(\omega^{2k}+\omega^k+1)$. If $k\equiv 0(\bmod 3)$, then $\omega^{2k}+\omega^k+1=3$. If k is not 0 mod 3, then $\omega^k\neq 1$, get $\omega^{2k}+\omega^k+1=0$. Hence

$$\sum_{k\equiv 0(\bmod 3)}\binom{n}{k}=\frac{1}{3}\sum_{k=0}^{n}\binom{n}{k}(1+\omega^k+\omega^{2k})$$
$$=\frac{1}{3}\sum_{k=0}^{n}\binom{n}{k}(1+\omega^k+\overline{\omega}^k)$$
$$=\frac{1}{3}(2^n+(1+\omega)^n+(1+\overline{\omega})^n)$$
$$=\frac{1}{3}(2^n+2\cos\frac{n\pi}{3}).$$

Note that $\omega^2=\overline{\omega}$ and $1+\omega=\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}$. By the same idea, we let $\varepsilon=e^{i\frac{2\pi}{m}}$ and then get a mod m selector:

$$\frac{1}{m}\sum_{k=0}^{m-1}e^{s(k-r)}=\begin{cases}1 & k\equiv r\bmod m\\ 0 & \text{otherwise}\end{cases}.$$

Example: Factorize $f(x)=x^{12}+x^9+x^6+x^3+1$ into product of two non-constant terms.

Let ε be any complex fifth root of unity. Then $f(\varepsilon)=\varepsilon^2+\varepsilon^4+\varepsilon^1+\varepsilon^3+1=0$, hence $x^4+x^3+x^2+x+1$ is a factor. By long division to get another factor $x^8-x^7+x^6-x^4+x^3-x+1$.

Example (IMO 1974 No. 3): Show that for any nonnegative integer n, the number $\sum_{k=0}^{n}\binom{2n+1}{2k+1}2^{3k}$ is not divisible by 5.

Because $2^3 \equiv -2 \pmod 5$, it suffices to show that $\sum_{k=0}^{n}\binom{2n+1}{2k+1}(-2)^k$ is not divisible by 5. Now expand $(1+i\sqrt{2})^{2n+1}$, get $(1+i\sqrt{2})^{2n+1} = R_n + i\sqrt{2}S_n$, where the real part $R_n = \sum_{k=0}^{n}\binom{2n+1}{2k}(-2)^k$ and the complex part $\sqrt{2}S_n = \sqrt{2}\sum_{k=0}^{n}\binom{2n+1}{2k+1}(-2)^k$. Multiplying the complex number with its conjugate, get $3^{2n+1} = R_n^2 + 2S_n^2$. By considering the equation mod 5, get $(-1)^n 3 \equiv R_n^2 + 2S_n^2 \pmod 5$. If $S_n \equiv 0 \pmod 5$, then $R_n^2 \equiv \pm 3 \pmod 5$, impossible.

<u>Example (IMO 1995, No. 6)</u>: Let p be an odd prime number. Find the number of subsets A of the set $\{1,2,...,2p\}$ such that (i) A has exactly p elements, and (ii) the sum of all the elements in A is divisible by p.

Let $\xi = \cos\frac{2\pi}{p} + i\sin\frac{2\pi}{p} = e^{i\frac{2\pi}{p}}$ be a primitive root of unity. Then $\prod_{i=1}^{2p}(x-\xi^i) = (x^p-1)^2 = x^{2p} - 2x^p + 1$. By comparing the coefficient of x^p, get

(1) $2 = \sum_{1\le i_1 < i_2 < ... < i_p \le 2p} \xi^{i_1+i_2+...+i_p} = \sum_{j=0}^{p-1} n_j \xi^j$, where n_j is the number of subsets $\{i_1, i_2, ..., i_p\}$ of $\{1,2,...,2p\}$ such that $i_1 + i_2 + ... + i_p \equiv j \pmod p$. In this case we look for n_0. Another observation: formula (1) also says that ξ is a root of the integer polynomial of degree $p-1$, $p(x) = n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j$. The minimal polynomial of ξ is $1 + x + x^2 + ... + x^{p-1}$. This means (2) $n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j = c(1 + x + x^2 + ... + x^{p-1})$. Put $x = 1$, get $n_0 - 2 + \sum_{j=1}^{p-1} n_j = cp$, hence $c = \frac{1}{p}(\sum_{j=0}^{p-1} n_j - 2) = \frac{1}{p}(\binom{2p}{p} - 2)$. Use c in (2) to get $n_o = 2 + \frac{1}{p}(\binom{2p}{p} - 2)$.
(Note: ξ is a primitive root means $\xi^p = 1$, but $\xi^n \ne 1$ for any $1 \le n < p$. ξ satisfies the equation $x^p - 1 = 0$ but not $x - 1 = 0$, hence it satisfies

$\phi(x) = x^{p-1} + x^{p-2} + \ldots + 1 = 0$. It can be shown that $\phi(x)$ is irreducible into polynomials of integer coefficients (Eisenstein's irreducibility criteria), hence $\phi(x)$ is a minimal polynomial of ξ. Any two minimal polynomials of ξ are rational multiples of the other.)

Applications to Trigonometry

Example: Show that $\cos^n \theta = \frac{1}{2^n} \sum_{k=0}^{n} \binom{n}{k} \cos((n-2k)\theta)$.

$$\cos^n \theta = \left(\frac{e^{i\theta} + e^{-i\theta}}{2} \right)^n$$

$$= \frac{1}{2^n} \sum_{k=0}^{n} \binom{n}{k} (e^{i\theta})^{n-k} (e^{-i\theta})^k$$

$$= \frac{1}{2^n} \sum_{k=0}^{n} \binom{n}{k} (e^{i(n-2k)\theta})$$

$$= \frac{1}{2^n} \sum_{k=0}^{n} \binom{n}{k} \cos((n-2k)\theta).$$

Example: Show that $\sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7} = \frac{\sqrt{7}}{2}$.

Let $\alpha = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7} = e^{i\frac{2\pi}{7}}$, $p = \alpha + \alpha^2 + \alpha^4$, and $q = \alpha^3 + \alpha^5 + \alpha^6$. Then by the De-Moivre's theorem $p + q = -1$ and $pq = 2$. Hence p and q are roots of the equation $x^2 + x + 2 = 0$. Roots of the equation are $\frac{-1 \pm \sqrt{-7}}{2} = \frac{-1 \pm \sqrt{7}i}{2}$. The required quantity is the imaginary part of $\frac{-1 + \sqrt{7}i}{2}$.

Example (IMO 1963, No. 5): Show that $\cos\frac{\pi}{7} - \cos\frac{2\pi}{7} + \cos\frac{3\pi}{7} = \frac{1}{2}$.

Let $z = \cos\frac{\pi}{7} + i\sin\frac{\pi}{7}$, then $z^7 + 1 = 0$. As $z \neq -1$ and
$z^7 + 1 = (z+1)(z^6 - z^5 + z^4 - z^3 + z^2 - z + 1) = 0$, it follows that
$z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$. Thus
$1 = -(z^6 - z^5 + z^4 - z^3 + z^2 - z) = -(z^3 - z^2 + z)(z^3 - 1)$, or $\frac{1}{1 - z^3} = (z^3 - z^2 + z)$. Now $\cos\frac{\pi}{7} - \cos\frac{2\pi}{7} + \cos\frac{3\pi}{7} = \mathrm{Re}(z^3 - z^2 + z)$, we have to prove $\mathrm{Re}(\frac{1}{1 - z^3}) = \frac{1}{2}$, which

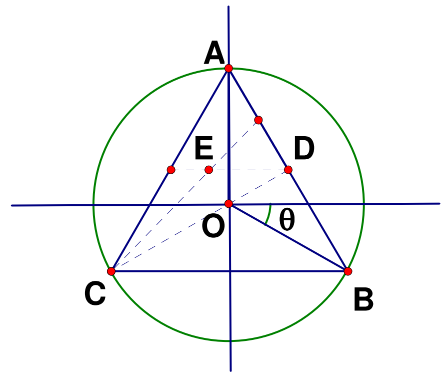
is clear.

(Indeed for any complex number $e^{i\theta}$ of modulus 1, we have

$$\frac{1}{1-e^{i\theta}}=\frac{1}{1-e^{i\theta}}\cdot\frac{1-e^{-i\theta}}{1-e^{-i\theta}}=\frac{1-\cos\theta+i\sin\theta}{2-2\cos\theta},$$ and its real part is $\frac{1-\cos\theta}{2-2\cos\theta}=\frac{1}{2}$.)

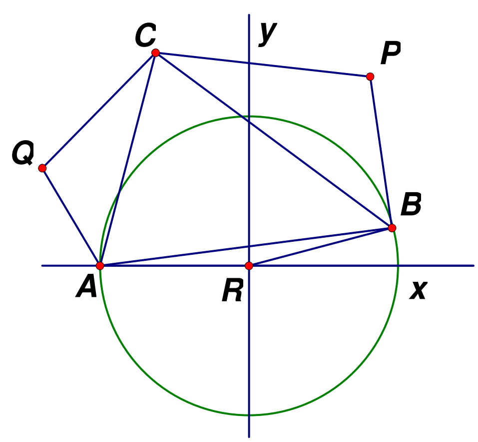
<u>Applications to Geometry</u>

<u>Example (BMO 1983)</u>: O is the circumcentre of ΔABC, D is the midpoint of AB, E is the centroid of ΔACD. If $AB=AC$, then $OE\perp CD$.



We make the origin the circumcentre and the circle is of radius 1 so that A corresponds to the complex number i ($z_A=i$). From the diagram $z_B=\cos\theta+i\sin\theta$, $z_C=-\cos\theta-i\sin\theta$. $\overrightarrow{OE}=\frac{1}{3}(z_A+z_C+z_D)=-\frac{1}{6}\cos\theta+\frac{1}{2}(1-\sin\theta)i$. And $\overrightarrow{CD}=z_D-z_C=\frac{3}{2}\cos\theta+\frac{1}{2}(1+\sin\theta)i$. Then show that $\frac{\overrightarrow{CD}}{\overrightarrow{OE}}$ is purely imaginary.

<u>Example (IMO 1975, No, 3)</u>: Let ABC be a triangle. The triangles ABR, BCP and CAQ are drawn externally on the sides AB, BC, CA respectively, such that $\angle PBC=\angle CAQ=45^\circ$, $\angle BCP=\angle QCA=30^\circ$ and $\angle RBA=\angle RAB=15^\circ$. Show that $\angle QRP=90^\circ$ and $QR=RP$.



Align the triangle in the plane so that $z_A=-1$ and R be the origin (see diagram).

We have to prove $z_Q=iz_P$. Indeed $z_A=-1$ hence $z_B=\cos 30^\circ+i\sin 30^\circ$ (isosceles triangle). Now $\frac{BP}{BC}=\frac{AQ}{AC}=\frac{\sin 30^\circ}{\sin 105^\circ}=\frac{\sqrt{2}}{1+\sqrt{3}}$, hence

$$z_P=z_B+\overrightarrow{BP}=z_B+\frac{\sqrt{2}}{1+\sqrt{3}}(z_C-z_B)e^{-i\frac{\pi}{4}},$$

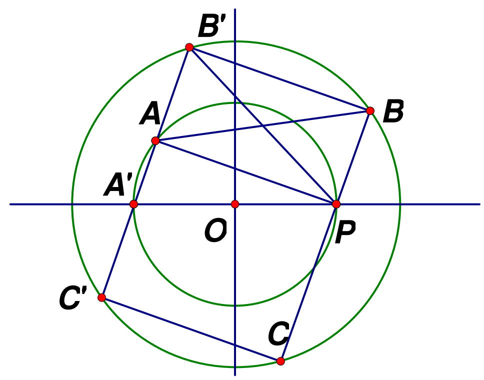
and

$z_Q = z_A + \overrightarrow{AQ} = z_A + \frac{\sqrt{2}}{1+\sqrt{3}}(z_C - z_A)e^{i\frac{\pi}{4}}$. It is routine to check $z_Q = iz_P$.

<u>Example (IMO 1988, No, 1)</u>: Consider two coplanar circles of radii R and r ($R > r$) with the same centre. Let P be a fixed point on the smaller circle and B a variable point on the larger circle. The line BP meets the larger circle again at C. The perpendicular l to BP at P meets the smaller circle again at A (if l is tangent to the circle at P then $A = P$).

Find the set of values of $BC^2 + CA^2 + AB^2$;

Find the locus of the midpoint of AB.



In the diagram $BCC'B'$ is a rectangle, with BC' meets the smaller circle at A and A' as shown. The distance from O to BC and $B'C'$ (AA') is the same. Hence $z_B + z_C + z_A + z_{A'} = 0$, or $z_B + z_C + z_A = -z_{A'}$. Now

$$\begin{aligned}
&BC^2 + CA^2 + AB^2 \\
&= (z_C - z_B)(\overline{z_C} - \overline{z_B}) + (z_A - z_C)(\overline{z_A} - \overline{z_C}) + (z_B - z_A)(\overline{z_B} - \overline{z_A}) \\
&= 4R^2 + 2r^2 - (z_B\overline{z_A} + z_A\overline{z_B} + z_C\overline{z_B} + z_B\overline{z_C} + z_B\overline{z_C} + z_C\overline{z_B}) \\
&= 6R^2 + 3r^2 - (z_A + z_B + z_C)(\overline{z_A} + \overline{z_B} + \overline{z_C}) \\
&= 6R^2 + 2r^2.
\end{aligned}$$

Easy to check, the result is still true when A and P overlaps.

Since $BB'AP$ is also a rectangle, the midpoint of AB is also the midpoint of $B'P$, and let it be Q. Then $z_{B'} + z_P = 2z_Q$, hence $\left|z_Q - \frac{z_P}{2}\right| = \left|\frac{z_{B'}}{2}\right| = \frac{R}{2}$, locus is the circle with centre the midpoint of OP, and radius $\frac{R}{2}$.

Exercises (Complex Numbers)

1. Three nonzero complex numbers z_1, z_2, z_3 satisfying $z_1 + z_2 + z_3 = 0$ and $|z_1| = |z_2| = |z_3|$. Show that they correspond to the vertices of an equilateral triangle.

2. (CMO 1986, No.2) Let $z_1, z_2, ..., z_n$ be complex numbers satisfying $|z_1| + |z_2| + ... + |z_n| = 1$. Show that the modulus of some of the members in this set is not less than $\frac{1}{6}$.

3. Given 8 nonzero real numbers $a_1, a_2, ..., a_8$, show that at least one of the 6 numbers $a_1a_3 + a_2a_4$, $a_1a_5 + a_2a_6$, $a_1a_7 + a_2a_8$, $a_3a_5 + a_4a_6$, $a_3a_7 + a_4a_8$, $a_5a_7 + a_6a_8$ is nonnegative.

4. Suppose $\cos\alpha + \cos\beta + \cos\gamma = \sin\alpha + \sin\beta + \sin\gamma = 0$, show that $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = \sin 2\alpha + \sin 2\beta + \sin 2\gamma = 0.$

5. Let x and y be nonzero complex numbers satisfying $x^2 + xy + y^2 = 0$, find $\left(\frac{x}{x+y}\right)^{1990} + \left(\frac{y}{x+y}\right)^{1990.}$

6. Evaluate $\sum_{k \equiv 1 \pmod 3} \binom{n}{k}$ and $\sum_{k \equiv 2 \pmod 3} \binom{n}{k}$.

7. Suppose $3z^6 + 2iz^5 - 2z - 3i = 0$, find $|z|$.

8. Suppose $|z| = 1$, find the maximum possible value of $u = |z^3 - 3z + 2|$.

9. Let a, b, x and y be real numbers satisfying $x^2 + y^2 \le 1$ and $a^2 + b^2 \le 2$. Show that $|b(x^2 - y^2) + 2axy| \le \sqrt{2}$.

10. Let $|z| \le 1$ and $|w| \le 1$, show that $|z + w| \le |1 + \bar{z}w|$.

11. (IMO 1990) Prove that there exist a convex 1990-gon with the following properties:
    (a) all angles are equal,
    (b) the lengths of the sides are the numbers $1^2, 2^2, 3^2, ..., 1989^2, 1990^2$ in some order.

Gaussian Integers

The set $R = \mathbb{Z}[i] = \{x + yi \mid x, y \in \mathbb{Z}\}$ is the set of *Gaussian Integers*, where $i^2 = -1$.

Define the *Norm* function such that $N(x+yi) = x^2 + y^2$. If $a = x + yi$, define its *conjugate* $\bar{a} = x - yi$. Hence $N(x+yi) = x^2 + y^2 = a\bar{a}$. Check that for (i) $\overline{ab} = \bar{a}\bar{b}$, hence (ii) $N(ab) = N(a)N(b)$. Also a is a *unit* if $N(a) = 1$. Most important of all, $R = \mathbb{Z}[i]$ is a *Euclidean Ring*, namely one has the Euclidean Algorithm as ordinary integers:

Theorem. $R = \mathbb{Z}[i]$ is a Euclidean Ring.

Let $a, b \neq 0$, write $\frac{a}{b} = p + iq$, with $p, q \in \mathbb{Q}$. Let m, n be integers such that $|m - p| \leq \frac{1}{2}$ and $|n - q| \leq \frac{1}{2}$. Thus $a = b(m + ni) + r$, with $a - b(m + ni) = r$. Now

$$\begin{aligned}
&N(r) = N(ab^{-1} - m - ni)b) \\
&= N(p + qi - m - ni)N(b) \\
&= N((p - m) + (q - n)i)N(b) \\
&\leq (\frac{1}{4} + \frac{1}{4})N(b) = \frac{N(b)}{2} < N(b).
\end{aligned}$$

As in integers, we can apply the rule iteratively to obtain the Euclidean algorithm.

Remarks.

(i) We define $a|b$ (a *divides* b) if $b = ac$.

(ii) u is unit if and only if u divides 1. Indeed if u is a unit, then $N(u) = u\bar{u} = 1$, hence u divides 1. Conversely if $u|1$, then $1 = uv$. Taking norms, get $N(u) = N(v) = 1$.

(iii) u is a *prime* if $u|ab$, then $u|a$ or $u|b$.

(iv) v (a non-unit) is *irreducible* if $v = ab$ implies either a or b a unit.

(v) Only units in $R = \mathbb{Z}[i]$ are $1, -1, i, -i$.

In the Gaussian Integers, irreducibles and primes are the same. Clearly a prime is irreducible (reducible cannot be primes). Conversely suppose v is irreducible and $v|ab$, and $v \nmid a$, then v and a are relatively prime. Thus we can find c and d such that $cv+da=1,$ hence $cvb+dab=b.$ This implies $v|b.$

Hence we can reduce an element of $\mathbb{Z}[i]$ into as many irreducibles (primes) as possible and show that

Theorem. $\mathbb{Z}[i]$ is a *unique factorization domain*.

(*Primes* in $\mathbb{Z}[i]$ are called *Gaussian Primes*.

Exercises.

1. Check if $a|b$?

(a) $a=3, b=4+7i$ (No, if $a|b$, then $N(a)|N(b)$. (b) $a=5+3i$, $b=30+6i$ (No, same as a). (c) $a=2+i$, $b=15$ (yes, quotient is $6+3i$). (d) $a=11+4i$, $b=274$ (yes, quotient is $22-8i$).

2. Two Gaussian integers *a* and *b* are *associates* if $a=bu$, with *u* a unit. When are $a+bi$ and associate of $a-bi$? (Try 4 possible units to get real numbers, purely imaginary numbers, $a+ai$ or $a-ai$).

3. All Gaussian Integers of norm ≤ 35 divisible by $4-i.$ (Multiples are of the form $(4-i)(x+yi)=(4x+y)+(5y-x)i.$ Take norms to observe $N(x+yi)\le 2,$ get $x+yi=0,\pm 1,\pm i,\pm 1\pm i$).

4. Find GCD.
   (a) $a=24-9i, b=3+3i$ $(-3i)$.
   (b) $a=87i, b=11-2i$ (1).
   (c) $a=18+15i, b=3+4i$ (1).
   (d) $a=-2+8i, b=5-i$ $(1\pm i)$.
   (e) $a=13+7i, b=2+3i$ (relatively prime).

5. If $a|b$, then $N(a)|N(b)$.

6. If $a+bi$ is a nonzero Gaussian Integer, then it has exactly one associate of the form $c+di$, with $c>0, d\ge 0$. (Try cases).

7. For every pair of Gaussian Integers a and b, with $0\ne b\nmid a$, there are at least two different c and d, such that $a=bc+d$, with $N(d)<N(b)$. (Problem of Rounding).

8. Find all pairs (a,b) such that $a^2+b^2=650$.

We now ask when is an (ordinary) prime a sum of two squares. . Indeed

Lemma. If $p=2$ or p a prime and such that $p \equiv 1 \pmod 4$, then $x^2+1 \equiv 0 \pmod p$ has a solution in integers.

Proof. For $p=2$, it is clear. Now let $p=4k+1$, then the integers $1,2,\ldots,2k,2k+1,\ldots,4k$ are congruent to $1,2,\ldots,2k,-2k,-(2k-1),\ldots,-1$ mod p. By Wilson,
$-1 \equiv (1)(2)\ldots(2k)(-2k)(-(2k-1))\ldots(-1) = (-1)^2((1)(2)\ldots(2k))^2 = ((1)(2)\ldots(2k))^2$ mod p, and hence the result.

Theorem. Given a prime p, it is the sum of two squares if and only if $p=2$ or $p \equiv 1 \pmod 4$.
Proof. If $p \equiv 3 \pmod 4$, then $p = x^2+y^2$ is impossible, simply by considering the equation modulo 4. Also $2 = 1^2+1^2$. Suppose now $p \equiv 1 \pmod 4$, then we find $c, x\, y$ such that $cp = x^2+1 = (x+i)(x-i)$. Now suppose p is irreducible (prime) in $\mathbb{Z}[i]$. Then p divides $x+i$ ot $x-i$, impossible. Hence p is reducible. The implies $p = uv$, a product of two units. Taking norms, get $p^2 = N(u)N(v)$, forcing $p = N(u) = N(v)$. Hence a solution of $p = a^2+b^2$.

Theorem. The primes of $\mathbb{Z}[i]$ are of the following types:
1. $1+i$,
2. integer primes $p \equiv 3 \pmod 4$,
3. factors $x \pm yi$ of integer primes $p \equiv 1 \pmod 4$,

and their associates.

Proof. The proof may be divided into several steps.
First if $N(a)$ is a rational (ordinary) prime, then a is a rational prime. Indeed if $a = bc$, then $N(a) = N(b)N(c)$. But $N(a)$ is prime, so one of b or c is a unit, so a is a rational prime.
Now we want to show a Gaussian prime divides a rational prime. Indeed u is a Gaussian prime, then $u\bar{u} = N(u) = n$. Hence $u \mid n = p_1 p_2 \ldots p_l$. This forces $u \mid p_i$, for some i.
Also $2 = (1+i)(1-i)$. Each of $1 \pm i$ is of norm 2, hence is a Gaussian prime.
Now suppose p is a prime such that $p \equiv 3 \pmod 4$. Suppose $p = ab$ in $\mathbb{Z}[i]$. Then $p^2 = N(p) = N(a)N(b)$. But $N(a)$ and $N(b)$, as sum of squares, so can only be $\equiv 0,1,2 \pmod 4$, impossible. Hence these primes remain Gaussian primes.
Finally if $p \equiv 1 \pmod 4$, then $p = x^2+y^2 = (x+yi)(x-yi)$. Each of $x \pm yi$ are of norm p, hence are Gaussian primes.

Exercises.

1. $7\,|\,(8-i)(4+5i)$? (No, take norms).

2, True or false? If $N(a)\,|\,N(b)$, then $a\,|\,b$? (No, take $a=2+i, b=2-i$).

3. Gaussian Primes? $3+4i$, $3-4i$, $11i$? (No, No, Yes, $2+i$).

4. Factorize $60+105i$ as Gaussian Primes. ($3(2+i)^2(2-i)(3+2i)$).

5. Factorize $239+i$ as Gaussian Primes. ($(1-i)(3+2i)^4$).

6. Gaussian Integers of norm 1800? How many of them? (Basically solve $u^2+v^2=1800=2^3 3^2 5^2$. $4(2+1)=12$ of them).

7. All Gaussian Integers of norm 169. (Solve $(u+vi)(u-vi)=169$, get $169=(2+3i)^2(2-3i)^2$. Thus $u+vi=\varepsilon(2+3i)^k(2-3i)^{2-k}$, where ε is a unit and $0\le k\le 2$. Take various units to get $\pm5\pm12i, \pm12\pm5i, \pm13, \pm13i$).

8. Find all Gaussian Primes of norm ≤ 60.

9. Solve $x+y+z=xyz=1$ in Gaussian Integers. (All of x,y,z are units, $(1,i,-i)$ and its permutations).

10. Factorize $x^2+y^2=(x+yi)(x-yi)$ to show that any primitive Pythagorean triple (x,y,z) with y even is of the form $x=a^2-b^2, y=2ab$ and $z=a^2+b^2$. (Note $x+yi$ and $x-yi$ are relatively prime).

11. There exist infinitely many rational primes of the form $4k+1$.

12. There exist infinitely many Gaussian primes of the form $a+bi$.

13. Let $n=2^e p_1^{e_1}p_2^{e_2}...p_k^{e_k}q_1^{f_1}q_2^{f_2}...q_l^{f_l}$ be the prime, factorization of n, p_i primes and $p_i\equiv 1 \pmod 4$, q_j primes and $q_j\equiv 3 \pmod 4$, with f_j even. Then the number of ordered pair (x,y), with $n=x^2+y^2$ (sum of two squares) is $4\prod_{i=1}^{k}(e_i+1)$. Other cases of n not possible. Check (i) $n=30420$, $4\times(1+1)(2+1)=24$, (ii) 650 (24).

14. Can you go from a Gaussian Prime, jumping with a bounded distance, to another Gaussian Prime, then go to infinity? (Not known, impossible in integers).

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